



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

CLIMATIC TEST CHAMBER THERMAL CYCLIC TEST CHAMBER



Sri Easwari Scientific Solution Pvt Ltd.

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SRI EASWARI SCIENTIFIC SOLUTION PVT LTD

Sri Easwari Scientific Solution Pvt Ltd (SESS) chennai is a leading name in the design and manufacturing of environmental test chambers. We are proud to be trusted by prestigious institutions and organizations such as ISRO, DRDO, IITs, and NITs. Our high-quality, innovative testing solutions are engineered to meet the rigorous demands of various sectors, ensuring precision, reliability, and efficiency.

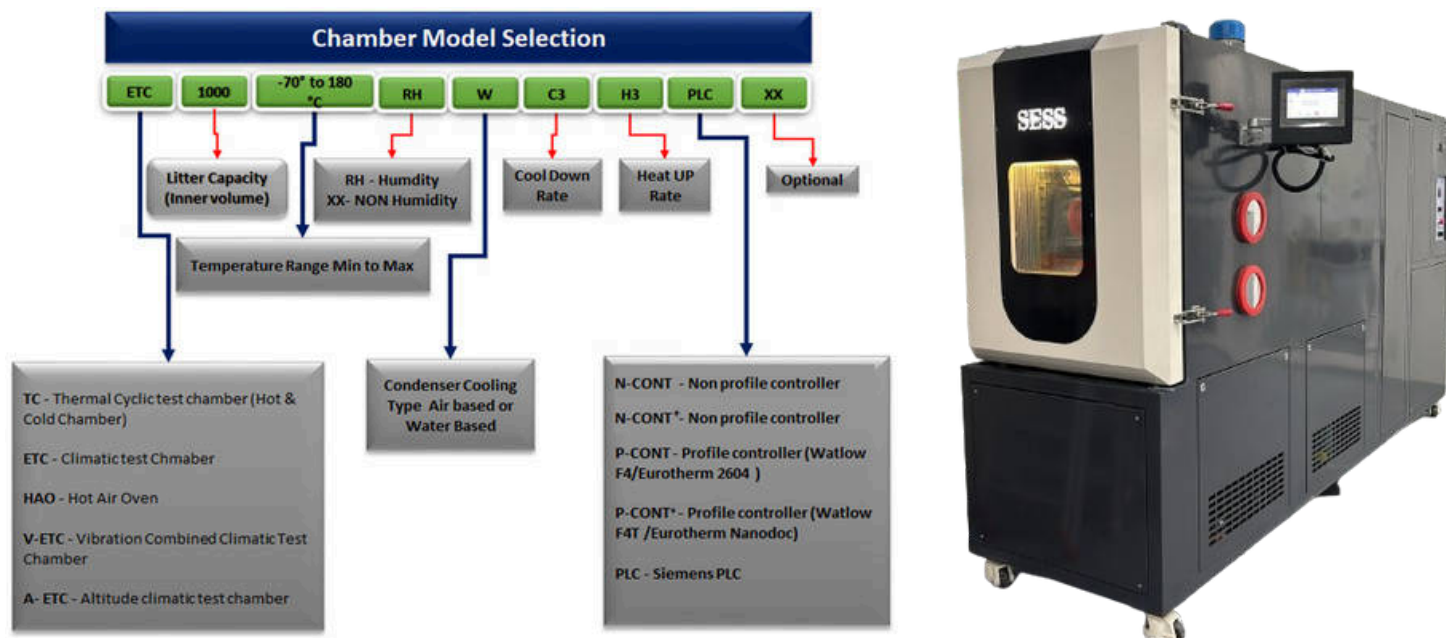


Our Key Products

- Climatic Test Chambers
- Thermal Shock Test Chambers
- Salt Spray Test Chambers
- Rapid Temperature Cycling Chambers
- Burn-In Test Chambers
- Solar Radiation Test Chambers

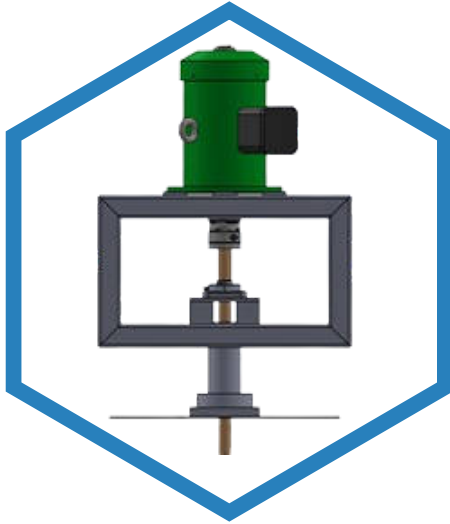


THERMAL CYCLIC TEST CHAMBER / CLIMATIC TEST CHAMBER ETC - XXXX/XXX TO XXX /XX/X/CX/HX/XXX/XX



Stand Model litter capacity
 Thermal cyclic test chamber (TC),
 Climatic test Chamber (ETC),
 Altitude - Thermal cyclic test chamber (ATC),
 Altitude - Climatic test chamber (AETC),
 All dimensions in mm

Litter	W	D	H
22	300	250	300
30	350	250	350
64	400	400	400
100	500	400	600
180	580	450	750
250	600	535	750
340	600	800	700
380	600	800	800
500	800	650	950
600	850	730	900
750	800	800	950
800	1100	800	950
1000	1000	1000	1000
1200	1000	1130	1020
1500	1000	1500	1000
1800	1100	1600	1020
2000	1000	2000	1020
2250	1100	2000	1020



In our climatic test chamber's air circulation system, we incorporate the following components and features:

- 1. Fan/Blower:** The chamber is equipped with a fan or blower featuring a continuous-duty rated motor. This motor efficiently circulates air within the chamber.
 - 2. Conditioning Space:** Inside the main chamber, there's a dedicated conditioning area with baffles, housing essential components like heaters and cooling coils. These elements play a crucial role in air conditioning.
 - 3. Uniform Temperature:** Conditioned air is introduced into the main chamber to ensure consistent temperature distribution throughout the workspace, maintaining a stable temperature environment during tests.
 - 4. Laminar Airflow:** The chamber is designed for laminar airflow, achieved using a fan motor and impeller with an appropriate capacity. This setup guarantees even distribution of temperature and humidity within the chamber, adhering to specified limits.
 - 5. Motor Shaft:** TC/ETC - The motor shaft used for chamber air circulation is constructed with SS 304 without any joints. The motor is positioned outside the chamber, while an extended-length SS 304 shaft drives the fans. This design ensures smooth and reliable operation while preserving the chamber's internal environment.
- ATC/AETC** - "The mechanical design of the chamber's air circulation motor is constructed with a sturdy shaft connected by a flexible coupling. The inner shaft is made of SS 304 material and is connected to the blower drum, which features a double vacuum seal mechanism. The motor is positioned outside the chamber, and it drives the blower drum through the inner SS 304 shaft. This design ensures smooth and reliable operation while maintaining the integrity of the chamber's internal environment.
- 6. Motor Winding:** For high-temperature chambers, we utilize motor windings of H class, whereas for chambers operating below 130°C, F class motor windings are employed.

These features combine to create an efficient and precise air circulation system within the climatic test chamber, facilitating accurate and dependable test conditions for various experiments and applications.

INCONEL-TYPE AIR HEATERS WITH PTFE WIRE AND SILICON END CAP



Heating System:

The chamber is equipped with sealed Inconel tube heaters to maintain a uniform temperature. These heaters are meticulously designed to ensure even heat distribution throughout the chamber. Heat input is regulated by solid-state relays, controlled through a PID controller programmer.

One notable advantage of inconel tube heaters is their exceptional ability to withstand high temperatures and corrosive environments. Inconel, known for its outstanding durability and heat resistance, is well-suited for prolonged use in challenging conditions.

Compared to stainless steel heaters, inconel heaters exhibit superior resistance to oxidation, corrosion, and high-temperature degradation. They are engineered to deliver reliable and efficient heat transfer without direct radiation onto the test item, ensuring a uniform heating process that doesn't adversely affect the test specimen.

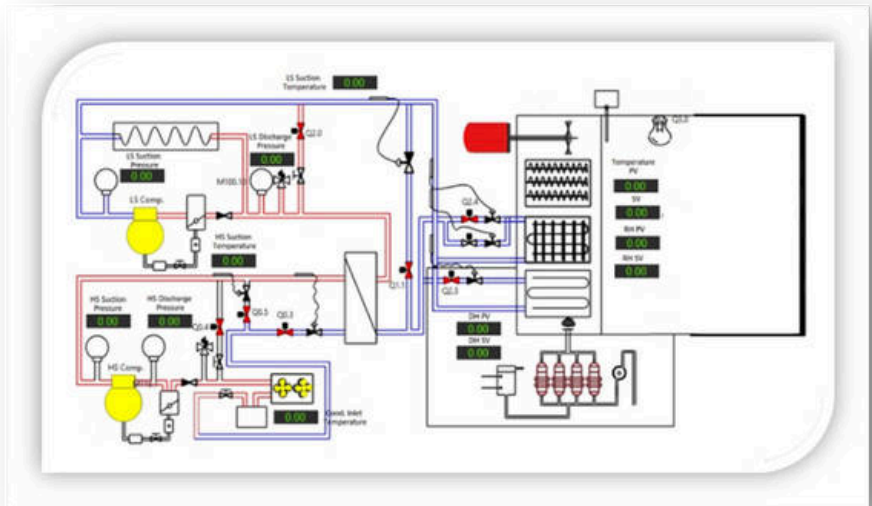
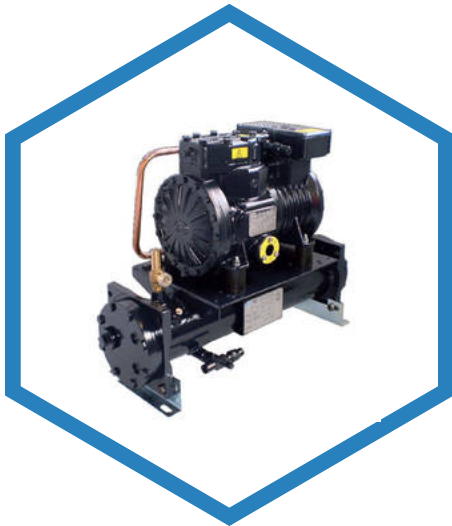
In conclusion, the use of sealed inconel tube heaters is highly recommended for maintaining precise and uniform chamber temperatures while ensuring the long-term performance and durability of the heating system.



Refrigeration System Based on Model Selected:

For single-stage configurations, the system features either air or water-cooled refrigeration, incorporating a Hermetic compressor from Embraco, Danfoss, or Emerson, or a semi-hermetic compressor from Bitzer, Dorin, or Bock. These compressors operate with the environmentally friendly refrigerant R404A. The system includes essential safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves, and is engineered for continuous and trouble-free operation.

For double-stage cascade configurations, the system comprises either air or water-cooled refrigeration, equipped with a Hermetic compressor from Embraco or Danfoss, or a semi-hermetic compressor from Bitzer, Dorin, Bock, or Emerson. These compressors utilize environmentally friendly refrigerants R404A and R23. Safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves are included, ensuring continuous and trouble-free operation.



Our chamber's refrigeration system is designed for optimal performance, with variations in specifications based on different models.

Single-stage compressor for temperatures up to -40°C , and a double-stage compressor for temperatures below -70°C . We select our compressors from reputable brands like Bitzer, Dorin, Bock, Danfoss, SECOP, or Embraco.

Environmentally friendly refrigerants are used, with R404A for single-compressor systems and R404A/R23 for cascade systems.

Safety features include HP/LP cut-off switches, pressure transmitters for continuous monitoring, and suction line temperature sensors for enhanced safety.

The system can have a water or air-cooled condenser based on the model. Temperature monitoring and safety limits are integrated into our PLC system for effective control and protection.

The compressor and condenser unit are positioned below the chamber for optimal performance.

For condenser cooling, our PLC system monitors the air or water inlet temperature. If it exceeds 26°C (water) or 40°C (air), a safety alarm triggers, and chamber operation stops until the temperature returns to normal, which can be adjusted through HMI and PC software. This setup ensures efficient cooling and temperature control for our thermal cyclic test chamber.



Control Panel & Wiring: A separate control panel will be provided, attached to the main chamber, which will house the programmer, on/off switches, fuses, contactors, indicating lamps, etc. Channel type wiring will be implemented using suitable copper wires with appropriate current ratings, along with marking ferrules, crimped dowel terminals, elmex connectors, and so on.

The electrical and instrumentation panel will be meticulously wired in compliance with international standards to ensure safety and reliability. Each electrical cable will be tagged with ID numbers, and the corresponding details will be clearly documented in the maintenance manual. Comprehensive circuit diagrams and relevant information will also be included.

To facilitate accurate and efficient design, we utilize SolidWorks 3D software for developing our drawings. This advanced software enables us to create detailed structural and electrical drawings that adhere to industry standards and specifications. By utilizing SolidWorks, we can ensure precise and comprehensive documentation of the electrical components and wiring layout of the Thermal cyclic test chamber.

Furthermore, we take pride in mentioning that our standard test chamber has been certified by TUV and carries the CE mark. This certification validates that our Stand test chamber only complies with the essential health, safety, and environmental protection requirements of the European Union. It serves as a testament to our commitment to providing high-quality, safe, and reliable equipment.

Our dedication to utilizing modern software tools and obtaining relevant certifications reflects our unwavering commitment to delivering superior products that meet the highest industry standards and exceed customer expectations.

Solid state relays for control for dry heater

TEST CHAMBER ALARM

In our all chamber indicators are available on the soft screen of the HMI itself, along with a buzzer option for the following:

- SPP trip
- Air Circulation Fan motor OLP
- Over Temperature in Chamber inner
- HS Compressor OLR
- HS Compressor over temperature
- HS Compressor HP trip
- HS Compressor LP trip
- LS Compressor OLR
- LS Compressor over temperature
- LS Compressor HP trip
- LS Compressor LP trip
- During transfer of basket (missing of the basket)
- R23 refrigerant low alarm
- HS R404A refrigerant low alarm
- ETC model - Water level Low
- Air Heater protection etc....

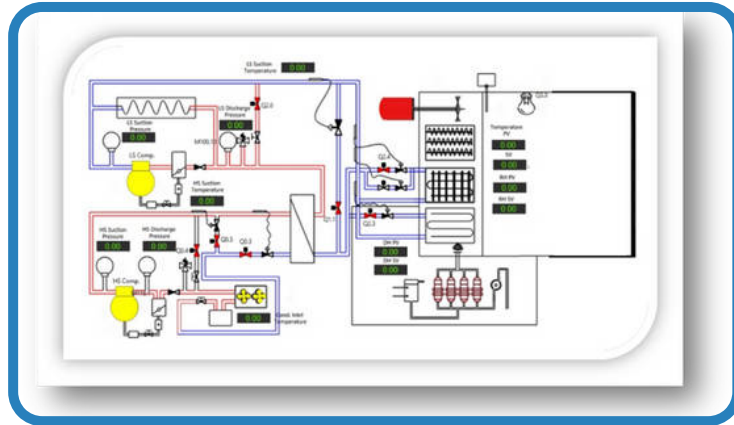


CONTROL SYSTEM



- Make: Siemens PLC: S7-1200 HMI: 7-inch Weintek Optional PC Software: SESS Climatic Test Viewer LabVIEW-based Software HMI Specifications:
- Internal Memory: 4 GB
- Data Logging: Up to 90 days, based on sampling duration (minimum 10 seconds)
- RAM Memory: 1 GB
- USB Port: 1
- Ethernet Ports: 2
- RS 485 Port: 1
- Data Log Files: Downloadable via USB and PC
- CMT Viewer: Accessible via PC and local network; mobile access through local network (customer-provided)
- Profiles: Up to 100 profiles, each with up to 100 segments (maximum 100 segments per profile)

In our chamber, the PLC (Programmable Logic Controller) will efficiently control and monitor the following aspects:



Temperature sensors:

The PLC will handle the set point temperature and continuously monitor the attained temperature values to ensure precise temperature control.

Status of compressor:

The PLC will monitor the operating status of the compressor, including start/stop commands and fault detection, to ensure proper functioning of the refrigeration system.

Pressure sensors:

The PLC will gather data from pressure sensors to monitor the pressure levels within the system and trigger appropriate actions based on predefined thresholds.

Valves:

The PLC will control various valves in the system, such as expansion valves, solenoid valves, and safety valves, to regulate the flow of refrigerant and other fluids.

Heater:

The PLC will control the operation of the heater, ensuring accurate and consistent heating as per the desired temperature profile.

Circulation Fan / Blower:

The PLC will regulate the speed and operation of the circulation fan or blower to maintain proper air circulation and temperature uniformity within the chamber.

Over current / over temperature protection: The PLC will incorporate protective measures, such as overcurrent and over temperature protection, to safeguard all the above components from potential damage or malfunctions.

By centralizing the control of these critical components, the PLC ensures efficient and reliable operation of the thermal cyclic test chamber while maintaining safety and accuracy in temperature control.

CONTROL SYSTEM PROFILE

Profile Monitor Profile Number: 1
PROFILE-TEST-1 **HOLD ON**

DURATION 000 : 15 : 00 ELAPSING 000 : 02 : 24

TEMPERATURE		HUMIDITY	
P.V	-40.2°C	P.V	000 %
S.V	000.0°C	S.V	000 %
Preset TEMP	010.0°C	Preset TEMP	000 %

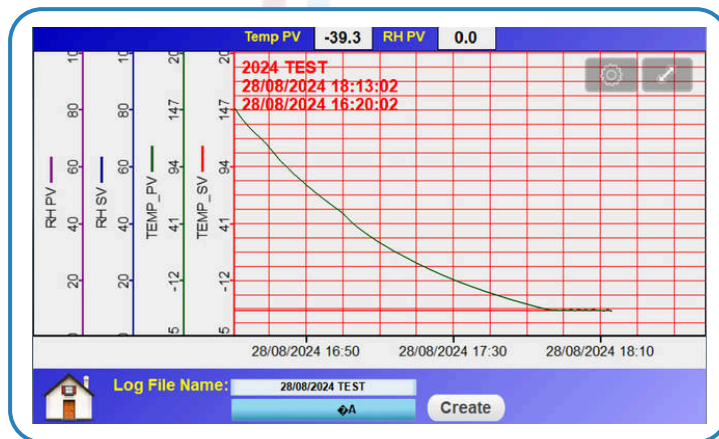
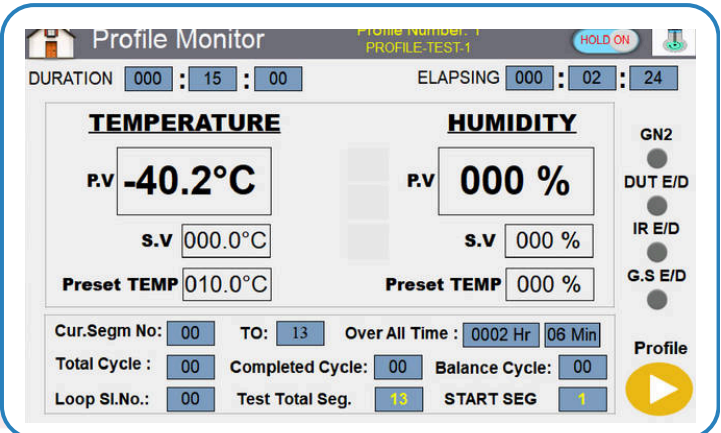
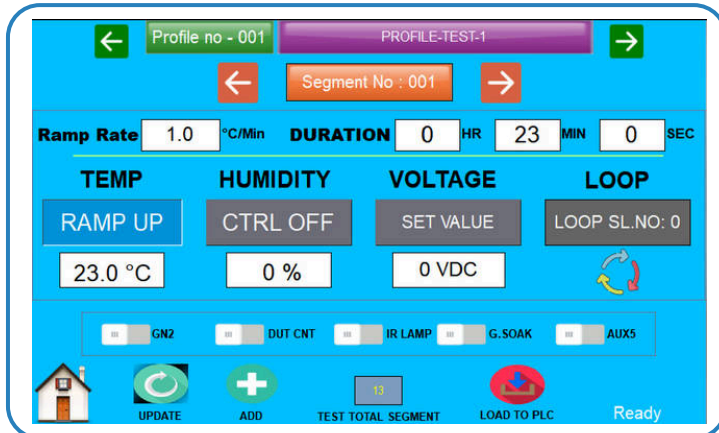
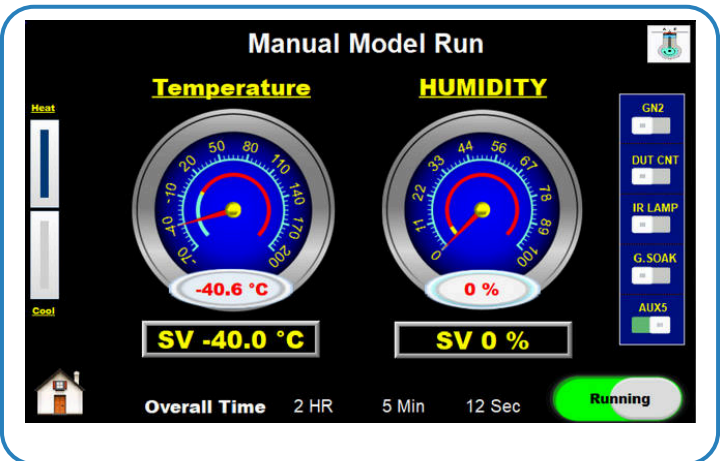
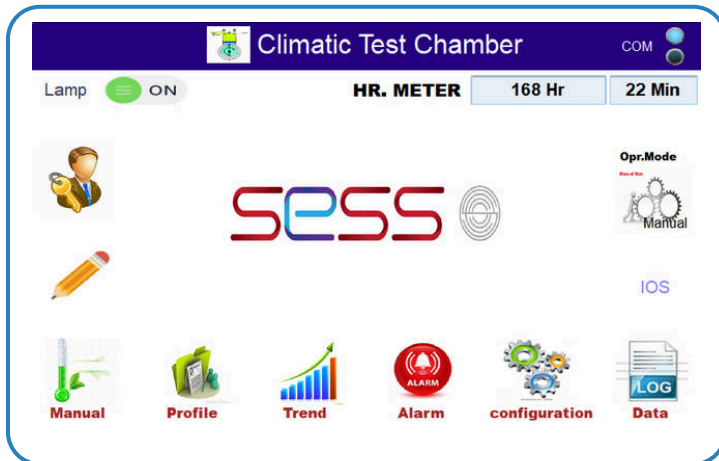
Cur.Segm No: 00 TO: 13 Over All Time : 0002 Hr 06 Min
 Total Cycle : 00 Completed Cycle: 00 Balance Cycle: 00
 Loop SI.No.: 00 Test Total Seg. 13 START SEG 1

GN2
DUT E/D
IR E/D
G.S E/D
Profile

- Programming: Up to 100 programs, each with 100 segments, repeatable up to 99 cycles
- In-segment looping
- Graphical program editing
- Auto PID configuration for precise temperature control
- Real-time trend monitoring
- Three levels of user access security
- Digital pressure monitoring for refrigeration compressor gas
- Monitoring of sub-system/component life cycles and run hours
- Calibration due warnings
- Individual compressor diagnostics: Oil level, Discharge temperature, Winding temperature, Suction temperature, run hour status with auto-report generation

The chamber is equipped with a feature to retain the system profile and program in the event of a power outage. Upon power restoration, the chamber seamlessly resumes operation from the exact point where it left off, ensuring uninterrupted continuity of operation. This feature is integral to our system, providing enhanced reliability and convenience for users.

HMI PAGE VIEWS



The 'Data Table' screen. It shows a table of test data. The table has columns for 'No', 'TIME', 'DATE', 'TEMP-SV', 'TEMP-PV', 'RH SV', 'RH PV', 'HS SUCT', 'HS DIS', 'LS SUCT', 'LS DIS', 'HS SUCT', and 'LS S'. The data is for the date '28/08/2024'. Below the table, there's a 'Log File Name' field with '28/08/2024 TEST' and a 'Create' button. There are also 'Logging' and 'Log Interval' settings.

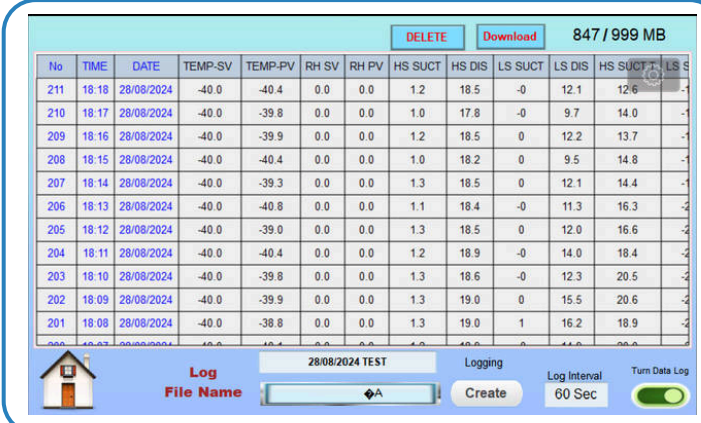
No	TIME	DATE	TEMP-SV	TEMP-PV	RH SV	RH PV	HS SUCT	HS DIS	LS SUCT	LS DIS	HS SUCT	LS S
211	18:18	28/08/2024	-40.0	-40.4	0.0	0.0	1.2	18.5	-0	12.1	12.6	-1
210	18:17	28/08/2024	-40.0	-39.8	0.0	0.0	1.0	17.8	-0	9.7	14.0	-1
209	18:16	28/08/2024	-40.0	-39.9	0.0	0.0	1.2	18.5	0	12.2	13.7	-1
208	18:15	28/08/2024	-40.0	-40.4	0.0	0.0	1.0	18.2	0	9.5	14.8	-1
207	18:14	28/08/2024	-40.0	-39.3	0.0	0.0	1.3	18.5	0	12.1	14.4	-1
206	18:13	28/08/2024	-40.0	-40.8	0.0	0.0	1.1	18.4	-0	11.3	16.3	-2
205	18:12	28/08/2024	-40.0	-39.0	0.0	0.0	1.3	18.5	0	12.0	16.6	-2
204	18:11	28/08/2024	-40.0	-40.4	0.0	0.0	1.2	18.9	-0	14.0	18.4	-2
203	18:10	28/08/2024	-40.0	-39.8	0.0	0.0	1.3	18.6	-0	12.3	20.5	-2
202	18:09	28/08/2024	-40.0	-39.9	0.0	0.0	1.3	19.0	0	15.5	20.6	-2
201	18:08	28/08/2024	-40.0	-38.8	0.0	0.0	1.3	19.0	1	16.2	18.9	-2

The 'Output List' screen. It shows a list of outputs and their status. The list is divided into two columns. The first column contains outputs like 'RELAY MAIN LINE', 'AIR CIRCULATION START', 'HS COMP START', etc. The second column contains outputs like 'LS COMP HOT GAS SOL', 'AIR_HEATER_SSR', 'STEAM_BOILER_HEATER_SSR', etc. Each output has a status indicator (ON/OFF).

Output Name	Status	Output Name	Status
RELAY MAIN LINE _ %Q0.0	OFF	LS COMP HOT GAS SOL _ %Q2.0	OFF
AIR CIRCULATION START _ %Q0.1	ON	AIR_HEATER_SSR _ %Q2.1	OFF
HS COMP START _ %Q0.2	ON	STEAM_BOILER_HEATER_SSR _ %Q2.2	OFF
HS COMP COOLING SOL _ %Q0.3	ON	DEHUMIDITY SOL _ %Q2.3	OFF
HS COMP HOT GAS SOL _ %Q0.4	OFF	STEAM WATER INLET PUMP _ %Q2.4	OFF
HS COMP INJECTION SOL _ %Q0.5	OFF	STEAM HEATER CONT _ %Q2.5	OFF
LS COMP START _ %Q0.6	ON	WINDOW HEATER _ %Q2.6	ON
LS COMP L COOLING SOL _ %Q0.7	OFF	INSPECTION LAMP _ %Q2.7	OFF
LS COMP S COOLING SOL _ %Q1.0	ON	ALARM BUZZER _ %Q3.0	OFF
LS COMP INJ SOL _ %Q1.1	OFF	MACHINE IDEAL CONDITIONS _ %Q3.1	OFF

HMI LOG DATA VIEWS

Data Storage and Profile Creation The Table Top Climatic Test Chamber boasts extensive data storage capabilities, allowing for the retention of test data for up to 100 days. Additionally, users have the ability to create and save up to 100 test profiles, streamlining the setup process and facilitating efficient testing procedures.



DELETED Download 847/999 MB

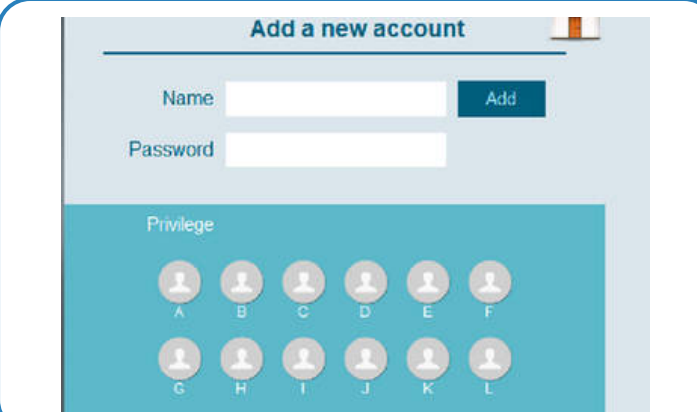
No	TIME	DATE	TEMP-SV	TEMP-PV	RH SV	RH PV	HS SUCT	HS DIS	LS SUCT	LS DIS	HS SUCT T	LS S
211	18:18	28/08/2024	-40.0	-40.4	0.0	0.0	1.2	18.5	-0	12.1	12.6	-1
210	18:17	28/08/2024	-40.0	-39.8	0.0	0.0	1.0	17.8	-0	9.7	14.0	-1
209	18:16	28/08/2024	-40.0	-39.9	0.0	0.0	1.2	18.5	0	12.2	13.7	-1
208	18:15	28/08/2024	-40.0	-40.4	0.0	0.0	1.0	18.2	0	9.5	14.8	-1
207	18:14	28/08/2024	-40.0	-39.3	0.0	0.0	1.3	18.5	0	12.1	14.4	-1
206	18:13	28/08/2024	-40.0	-40.8	0.0	0.0	1.1	18.4	-0	11.3	16.3	-2
205	18:12	28/08/2024	-40.0	-39.0	0.0	0.0	1.3	18.5	0	12.0	16.6	-2
204	18:11	28/08/2024	-40.0	-40.4	0.0	0.0	1.2	18.9	-0	14.0	18.4	-2
203	18:10	28/08/2024	-40.0	-39.8	0.0	0.0	1.3	18.6	-0	12.3	20.5	-2
202	18:09	28/08/2024	-40.0	-39.9	0.0	0.0	1.3	19.0	0	15.5	20.6	-2
201	18:08	28/08/2024	-40.0	-38.8	0.0	0.0	1.3	19.0	1	16.2	18.9	-2

Log File Name: 28/08/2024 TEST Logging: Log Interval: 60 Sec Turn Data Log: [ON]

PASSWORD PROTECTION

Certainly, our HMI provides comprehensive access level password protection. End-users will have their login IDs with customizable permissions based on different levels. The system allows for the creation of up to 12 levels of passwords, each with its specific permissions.

Furthermore, users with appropriate access levels can edit high/low alarm limits directly through the HMI interface. This functionality ensures that users can tailor alarm settings to their specific requirements without compromising security or ease of use.



Add a new account

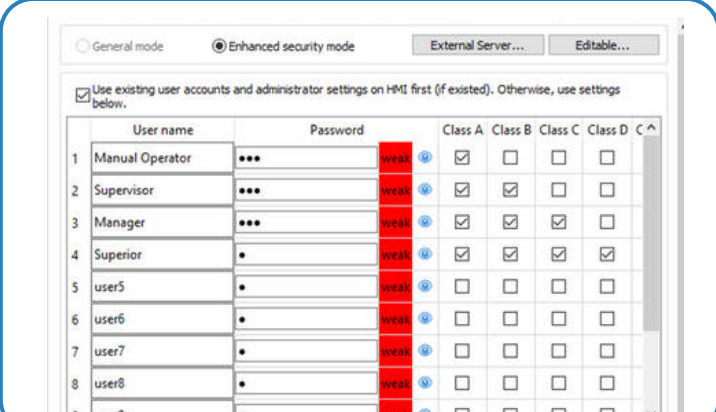
Name: Add

Password:

Privilege:

A B C D E F

G H I J K L



General mode Enhanced security mode External Server... Editable...

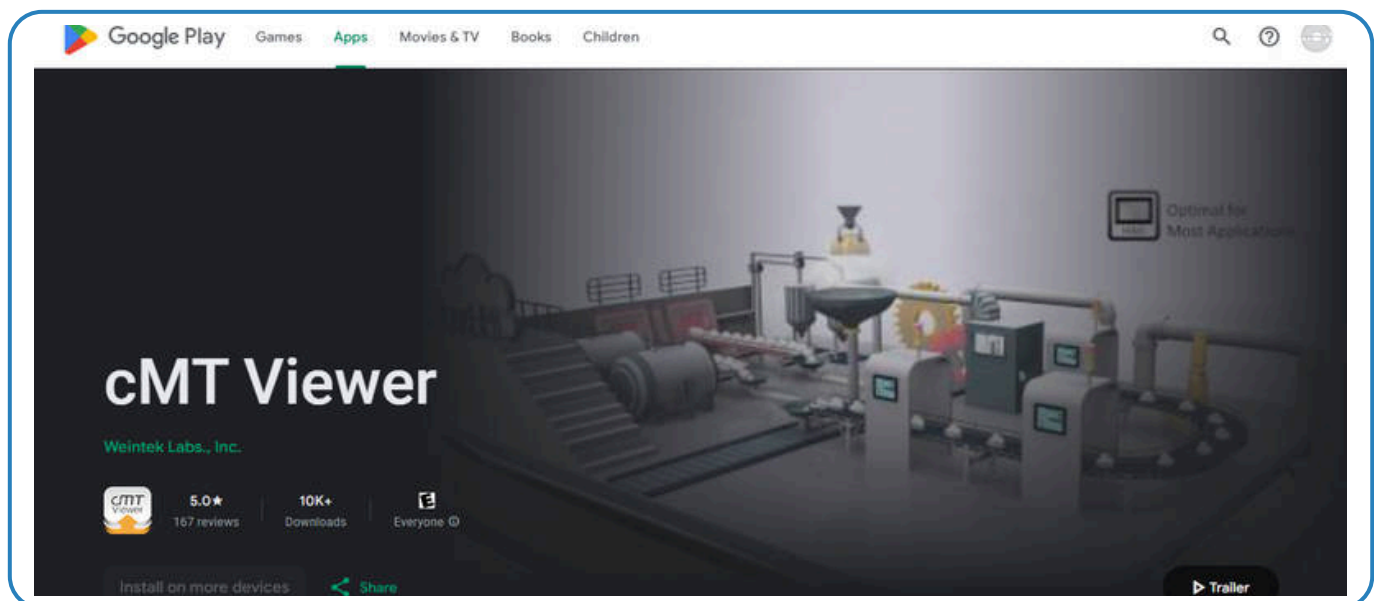
☒ Use existing user accounts and administrator settings on HMI first (if existed). Otherwise, use settings below.

User name	Password	Class A	Class B	Class C	Class D
1 Manual Operator	***	☒	☐	☐	☐
2 Supervisor	***	☒	☒	☐	☐
3 Manager	***	☒	☒	☒	☐
4 Superior	*	☒	☒	☒	☒
5 user5	*	☐	☐	☐	☐
6 user6	*	☐	☐	☐	☐
7 user7	*	☐	☐	☐	☐
8 user8	*	☐	☐	☐	☐

MULTI TEST CHAMBER CONTROL THROUGH REMOTE MONITORING AND CLOUD SUPPORT



CMT VIEWER IS AVAILABLE IN GOOGLE PLAY STORE



CERTIFICATION

ZERTIFIKAT ♦ CERTIFICATE ♦ 証書 ♦ CERTIFICADO ♦ CERTIFICAT



CERTIFICATE

The Certification Body
of TÜV SÜD South Asia Private Limited
certifies that

SESS

SRI EASWARI SCIENTIFIC SOLUTION PRIVATE LIMITED
2/298, ANE GARDEN, PERUMAL KOIL STREET,
SRINIVASAPURAM PARANIPUTHUR POST,
IYYAPPANTHANGAL CHENNAI - 600122, TAMILNADU, India

has implemented Quality Management System
in accordance with **ISO 9001:2015**
for the scope of

**DESIGN, MANUFACTURE, INSTALLATION, COMMISSIONING &
SERVICING OF INDUSTRIAL AND SCIENTIFIC EQUIPMENTS**

The certificate is valid from **2022-03-12** until **2025-03-11**
Subject to successful completion of annual periodic audits
The present status of this certificate can be obtained through TÜV SÜD website by scanning below QR code and by
entering the certificate number (without spaces) on web page. Further clarifications regarding the status & scope of
this certificate may be obtained by consulting the certification body at info@tuv-sud.com

Certificate Registration No. **99 100 19633**
Date of Initial certification: **2019-03-12**
Issue Date: **2022-02-15** Rev. 00




 Rahul Kale
Head of Certification Body
of TÜV SÜD South Asia Private Limited,
Mumbai
Member of TÜV SÜD Group





TÜV SÜD South Asia Pvt. Ltd. • TÜV SÜD House • Sakinaka • Andher (East) • Mumbai - 400072 • Maharashtra • India

TUVINDIA

Statement of Confirmation

No.: 2425-8119577562-002

Client's reference - Technical file number : SESS/CE/TCF/ETC/01- Rev.00

Name & address of the manufacturer : **M/s. SRI EASWARI SCIENTIFIC SOLUTION PRIVATE LIMITED**
Door No. 2/298, ANE Garden,
Perumal Kovil Street, Srinivasapuram,
Paraniputhur Post, Iyyappanthangal,
Chennai - 600 122,
Tamil Nadu, India

Product nomenclature : Environmental Test Chamber

Types : TC, ETC, Vibro TC, Vibro ETC, A-TC & A-ETC

Models : Refer to Annexure - I

Power Supply : 400 ± 10% V AC, 50 Hz, 3 Phase

Review results/observations

The Technical File referenced above submitted by the manufacturer has been reviewed for its document contents.
The Technical File generally covers the documentation content requirements of the European Directive(s):

- Machinery Directive : 2006/42/EC
- Electromagnetic Compatibility Directive : 2014/30/EU

Harmonised standard/s referred : EN 61010-1:2010/A1:2019, EN 61326-1:2013,
EN ISO 12100:2010

Other technical standard/s referred : -

Document issued by the manufacturer : EC Declaration of Conformity

Date of review : 09 August 2024

Jayprakash Hiremath
Vice President - Special Services

TUVIND/POCE/1501

Issue no:5
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TÜV

TUVNORDGROUP

THANK YOU!

Presented By
SESS TECHNICAL TEAM



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